

Trajecta: a model-driven approach for curricular analysis with Process Mining

Santiago Díaz^{1,*†}, Andrea Delgado^{1,†} and Daniel Calegari^{2,†}

¹*Instituto de computación, Facultad de Ingeniería, Universidad de la República, Julio Herrera y Reissig 565, 11300 Montevideo, Uruguay*

²*Universidad ORT Uruguay, Cuareim 1451, 11100 Montevideo, Uruguay*

Abstract

Curricular analysis aims to gain insight into the strengths and weaknesses of a curricular plan, identifying problems such as deviations in students' trajectories and bottlenecks, as well as evaluating the flexibility and constraints of the paths. Educational Process Mining (EPM) offers a process-centric perspective on curricular data. A key challenge for EPM is the availability of tools for curricular analysis that non-technical users can interpret, are open-source, and are readily available for use. In this paper, we present the Trajecta tool, which provides a model-driven approach for curricular analysis using Process Mining, benefiting both students and academic managers in managing curricular data. The tool offers a web interface, a Java component with a generic metamodel for modeling curricular programs, and a microservices Python component that leverages the PM4Py library to provide process mining functionalities. It was evaluated using an example application with real data from the Computer Science program at the Facultad de Ingeniería (FING), Universidad de la República (UdelaR).

Keywords

Educational Process Mining, Curricular analysis, Students' trajectories, Model Driven Engineering

Metadata description	Value
Tool name	Trajecta
Current version	1.2
Legal code license	GPL-3.0
Languages, tools and services used	Java, Python, Ecore, PM4Py, JavaScript, HTML, Flask, Servlets, choices.js, vis.js, Google Charts
Supported operating environment	Microsoft Windows, GNU/Linux - Requires Firefox, Chrome, or Edge internet browser
Download/Demo URL	https://gitlab.fing.edu.uy/open-coal/trajecta.git
Documentation URL	https://gitlab.fing.edu.uy/open-coal/trajecta
Source code repository	https://gitlab.fing.edu.uy/open-coal/trajecta
Screencast video	https://gitlab.fing.edu.uy/open-coal/trajecta/-/raw/main/demo.mp4

ICPM Doctoral Consortium and Demo Track 2025, October 20-24, 2025, Montevideo, Uruguay

*Corresponding author.

†These authors contributed equally.

✉ santiago.nicolas.diaz.conde@fing.edu.uy (S. Díaz); adelgado@fing.edu.uy (A. Delgado); calegari@ort.edu.uy (D. Calegari)

ORCID: 0009-0005-5626-5096 (S. Díaz); 0000-0003-4749-9366 (A. Delgado); 0000-0001-9506-7619 (D. Calegari)

 © 2025 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

1. Introduction

Curricular analysis [1] aims to gain insights into the strengths and weaknesses of a curricular plan, identifying problems such as students' trajectory deviations, curricular units that define bottlenecks and their impact, as well as evaluating paths' flexibility and constraints to detect improvement opportunities. Process Mining [2] plays a key role in this domain, as the curricular plans serve as a reference model for students to navigate their studies throughout their career, and the actual individual trajectories of students over this reference model define possible paths within it. Educational Process Mining (EPM) [3] provides such a view over curricular data.

In [4], the state of the art in Process Mining for curricular analysis is presented and discussed. One key specific challenge identified in that work concerns the availability of tools for curricular analysis that can be interpreted by non-technical users, i.e., tools that can be understood and used by domain experts. Although some tools exist such as [5, 6], they are not freely available for use by students and academic managers. Additionally, to support comparative studies between careers at different institutions and even across different countries, a common language is needed to model curricular plans and trajectories.

To close this gap, we propose the Trajecta tool, which provides support for curricular analysis for both students and academic managers, helping them deal with curricular data. The Trajecta tool offers a model-driven approach for modeling curricular programs, based on a metamodel that defines common yet generic concepts and relations for key elements, such as plans and courses, which can be used to model different careers from various institutions. It provides key functionalities, such as curricular trajectory path recommendations for students, and the application of key process mining analysis techniques, including discovering students' trajectory models, deviations analysis, bottlenecks, and data statistics for academic managers. The tool was evaluated using an example application with real data for the Computer Science career at the Facultad de Ingeniería, (UdelaR), validating its support for curricular modeling and analysis.

The rest of the document is organized as follows: Section 2 describes the functionalities of the tool, Section 3 introduces the architecture of the tool and technologies used, Section 4 discusses the maturity of the tool and finally Section 5 presents some conclusions and future work.

2. Trajecta Tool

The Trajecta tool follows a model-driven approach, which, at its core, is a metamodel for curricular programs, depicted in Figure 1, that allows modelling a wide range of study programs. It integrates key concepts, such as institutions and their careers, as well as concrete study plans for a career, i.e., a set of partially ordered curricular units that a student must complete. Moreover, it supports more dynamic concepts, such as instances of curricular units (i.e., courses over a given period), various types of periodic evaluations (partial, course, and exam), and students' enrollments in courses and assessments, along with their corresponding grades. A curricular plan is defined by creating a model that complies with the metamodel, serving as the reference model for the curricular plan and used for subsequent analysis and evaluation.

The tool provides two main categories of functionalities, each one designed for a specific user: i) curricular trajectory recommendation, which aims for students that want to know the

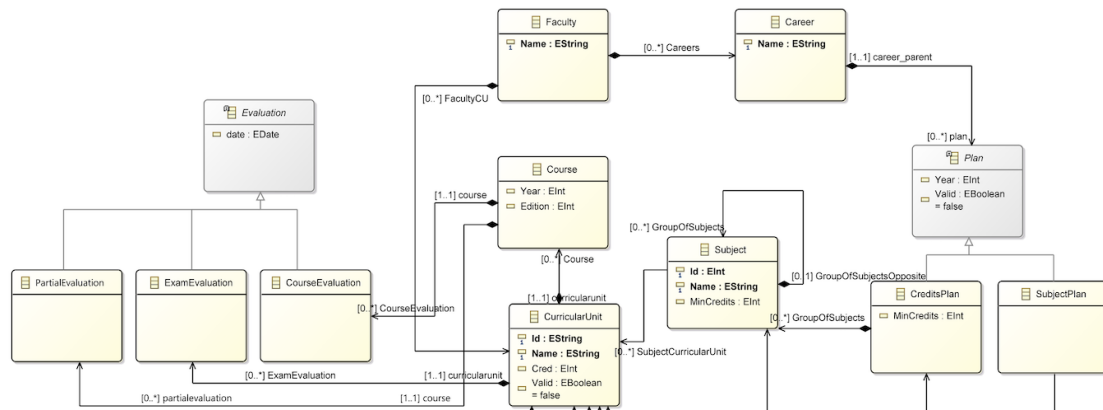
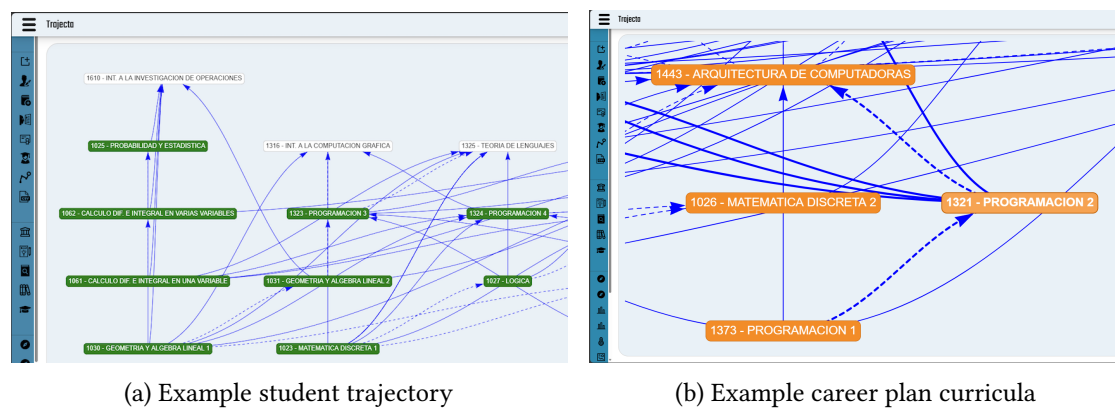


Figure 1: Generic curricular programs metamodel

available paths they can follow at any given moment, and ii) curricular trajectory analysis, which aims for academic managers that want to evaluate how students are traversing the curricula defined, where are the deviations, the bottlenecks, and different statics. In the following, these two categories and the main functionalities provided are introduced, illustrating the use of the tool with an example of application with real data from the Computer Science career at the Faculty of Engineering (UdelaR).

2.1. Curricular trajectory recommendation

Students can model their academic trajectory by entering information about their career, plan, courses, and evaluation results, which are then visualized as a graph. Figure 2a presents an example of a trajectory for a student of the Computer Science career at the Faculty of Engineering (UdelaR). The tool provides recommendations on curricular units that the student can enroll in, using a color-coded scheme: green for fully approved curricular units, yellow for partially approved curricular units, and white for available curricular units for enrollment.



(a) Example student trajectory

(b) Example career plan curricula

Figure 2: Curricular trajectory recommendation category functionalities

3. Architecture

The tool is presented as a web application, where the front-end can be deployed on any static server or service, such as Apache, GitHub Pages, GitLab Pages, or other services, or can be read directly from disk by opening the .html files. It utilizes *vis.js* to visualize the trajectory of a student and the curriculum of a career plan, and employs *choices.js* to enable searching within the select inputs of the tool. While the backend is divided into microservices, the process mining dedicated microservices are written in Python and use PM4Py [7], and the model-driven microservices are written in Java and use the Eclipse Modeling Framework (EMF)¹.

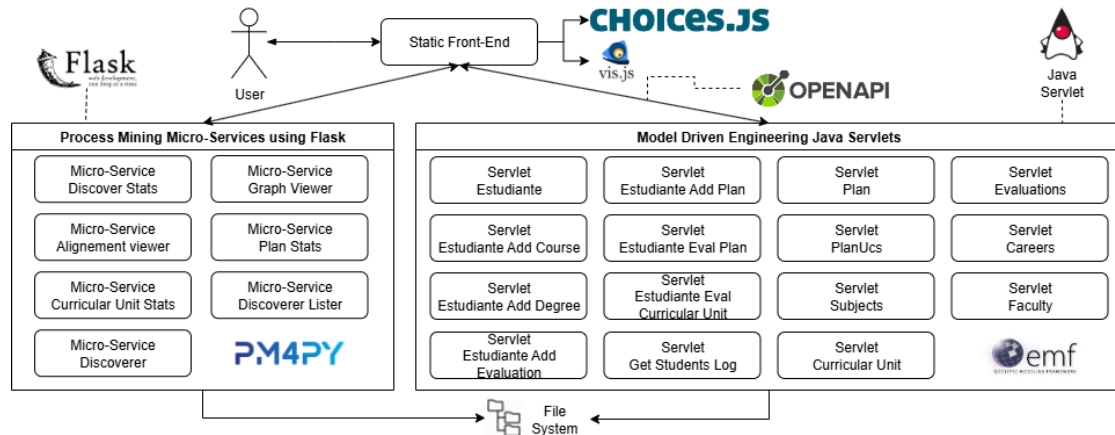


Figure 4: Tool Architecture with libraries used

The Java application provides functionalities for the category of curricula trajectory recommendation, including a metamodel with EMF support. The Python microservices support the process mining functionalities for the category of curricular trajectory analysis. The Java component also converts a student trajectory that is written in EMF to an event log. The Python component, as it is designed with microservices, allows for easy extension of functionalities, both those already provided by PM4Py and new ones that can be specifically developed.

4. Maturity of the tool

The Trajecta tool has been evaluated using an example application with real data from the Computer Science program at the Faculty of Engineering (UdelaR), as described in the previous sections. The model-driven approach was validated through the modeling of the curricular plan and the use of its functionalities by university students (with access to the front-end component and Java functionalities)². The process mining component, i.e., the PM4Py microservices backend, was validated using real data from the computer science career, specifically students' trajectories from 2018 to 2021. The event logs contain 2,297 cases (students) and 27,920 events, along with 1,708 variants, spanning from February 7, 2018, to March 5, 2021. Some performance

¹<https://eclipse.dev/emf/>

²<https://trajecta.fapret.com/>

limitations were identified during log discovery and graph visualization. The discoveries are executed in a background thread and do not notify the user when they are finished, which does not limit the tool's usage. Although initial usability and functional tests were conducted, primarily for the model-driven part of the tool by students, we plan to extend it to academic users for the process mining analysis component.

5. Conclusions

We have presented the Trajecta tool, which provides a model-driven approach for curricular analysis with Process Mining. The tool is open-source and available for free use, providing support for non-technical users and facilitating the modeling and analysis of different careers across various institutions, serving as a basis for comparative analysis. As part of our current and future work, we are integrating additional process mining functionalities to support specific educational analyses, with a focus on more user-friendly visualizations than those already provided. Additionally, we are evaluating the adaptability of the metamodel to model various types of study programs at different universities. This will also provide the basis for comparative process mining analysis between different institutions and countries.

Acknowledgments

We thank the Facultad de Ingeniería (FING), Universidad de la República (UdelaR) for providing the academic data and the Computer Engineering career curricula used in the demonstration. We also thank Santiago Freire, who contributed to the initial MDE-based development.

References

- [1] I. Hilliger, C. Aguirre, C. Miranda, S. Celis, M. Pérez-Sanagustín, Lessons learned from designing a curriculum analytics tool for improving student learning and program quality, *Journal of Computing in Higher Ed.* 3 (2022) 633–657.
- [2] W. M. P. van der Aalst, *Process Mining - Data Science in Action*, 2nd Ed., Springer, 2016.
- [3] A. H. Cairns, B. Gueni, M. Fhima, A. Cairns, S. David, N. Khelifa, P. Dautier, Process mining in the education domain, *Int. journal on advances in intelligent systems* 8 (2015) 219–232.
- [4] D. Calegari, A. Delgado, A systematic review on process mining for curricular analysis, *WIREs Data Mining and Knowledge Discovery* 15 (2025) e70005.
- [5] M. Pechenizkiy, N. Trcka, P. D. Bra, P. A. Toledo, CurriM: Curriculum mining, in: *5th Intl. Conf. on Educational Data Mining*, 2012, pp. 216–217.
- [6] M. Wagner, H. Helal, R. Roepke, S. Judel, J. Doveren, S. Görzen, P. Soudmand, G. Lakemeyer, U. Schroeder, W. M. P. van der Aalst, A combined approach of process mining and rule-based AI for study planning and monitoring in higher education, in: *Process Mining Workshops (ICPM 2022) Revised Selected Papers*, volume 468 of *LNBIP*, Springer, 2022, pp. 513–525.
- [7] A. Berti, S. van Zelst, W. van der Aalst, Process mining for python (pm4py): Bridging the gap between process-and data science, in: *ICPM Demo Track 2019*, volume 2374, *CEUR Workshop Proceedings*, 2019, pp. 13–16.